Jonathan Wesely

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CAREER OBJECTIVE

Computer Science B.S. and Applied Math B.S. Passionate about robotics, game/film dev, app/web dev, and automated workflows using AI agents. Knowledgeable in data structs/algos, OOP, probability, stats, differential equations, linear algebra, CAD, CAD/IDE-integrated AI tools, CI/CD, etc. I am currently working on projects in video game development using Unreal Engine 5, automation projects using AI agents, cross-platform mobile apps in Flutter (an AR golf swing analyzer and a Pokémon Champions battle companion), and merged MIPS-assembly contributions to the Smash Remix ROM hack. Most recently I taught physics and engineering at Merrill F. West High School's Space and Engineering Academy (2024–2026), where I also advised the VEX V5 Robotics Club. Available immediately for full-time work.

EDUCATION

University of California, Davis
Bachelor of Science: Computer Science | Bachelor of Science: Applied Mathematics | Graduated March 2023
Relevant coursework outside of majors: Physics: Mechanics, Thermodynamics, Electromagnetism, Modern Physics; Linux/Shell Scripting; Circuits I & II; Computer Architecture; General Chemistry

Purdue University (via Edureka)
9-month post-graduate program in DevOps (October 2023 – July 2024) | 6-month post-graduate program in AI & ML (October 2023 – April 2024)

Teachers College of San Joaquin: IMPACT Program
Single Subject Teaching Credential: Physics (October 2024 – June 2026)

ELVTR
"Multi-Agent AI For Game Development" course by Joshua Burdick (June 2026 – September 2026)

TECHNICAL SKILLS/TOOLS

Programming Languages: C/C++, Python, MATLAB, R, Dart
Mobile & Embedded: Flutter, Arduino, Bluetooth LE (custom GATT protocols), IMU sensor fusion, MIPS assembly (N64)
Markup Languages: HTML, LaTeX
Machine Learning Python Libraries: PyTorch, TensorFlow, Pandas, Keras, NumPy
IDEs: Cursor, Visual Studio, VS Code, repl.it, Jupyter Notebooks, Eclipse, Overleaf, Android Studio
Environment Management: Anaconda
Version Control: Git
Database Management: SQL, SQLite
Self-hosting: Docker, GitHub
CI/CD: GitHub Actions
Containerization: Docker
Operating Systems: Windows, Linux, Unix
Chat/Coding LLMs: Claude, Claude Code, Gemini, ChatGPT, Codex
Agentic Workflows: n8n, Claude Agent SDK, Claude Managed Agents
CAD: Meshy.ai, Copilot 3D, Blender, Maya, Onshape
Manufacturing Equipment: Sovol/Prusa/BambuLab 3D printers, FSL Muse CO_2 laser cutter
Game/Video Development Engines: Unreal Engine 5

PROJECTS

AR Golf Swing Analyzer (open source, in development) — *Flutter/Dart, Arduino, BLE, IMU sensor fusion*

Designing and building a ~10 g grip-mounted golf swing sensor (Arduino Nicla Sense ME + wide-range ICM-20649 IMU, solderless assembly) with a Flutter companion app. The on-phone processing pipeline computes club speed, face angle, and club path, validated against synthetic ground truth to ~0.15% speed and ~0.01° face-angle error. Includes a custom BLE GATT protocol with instant-metrics packets, automatic sensor failover, hands-free auto-arming, and multi-sensor support, with an AR heads-up display targeted for Snap Spectacles.

Pokémon Champions Battle Companion (in development) — *Flutter/Dart, SQLite, multimodal LLM vision*
Building a real-time battle-intelligence app for Pokémon Champions: photograph or screenshot the battle screen and an LLM-based recognizer identifies every on-field Pokémon. Features an SP-native team builder with per-regulation legality linting and a prediction engine that surfaces likely enemy movesets, items, abilities, and stat ranges from competitive usage data, including speed-tier comparisons. Mock-first architecture with ground-truth-tested stat math.

Smash Remix — Super Smash Bros. 64 ROM hack (contributor) — *MIPS assembly*
Shipped merged contributions to the community Super Smash Bros. 64 ROM hack, including a Tournament mode and move-buffering improvements, written in MIPS assembly and running on original N64 hardware.

AI Agents to Automate Processes at Work — *Claude Code, n8n, Claude Agent SDK, Claude Managed Agents*
Created agentic workflows that have automated many redundant teaching processes, including automated grading, automated Excel sheets for grades/budgets, automated lesson planning, and automated creation of instructional documents.

Automation of Detection of Bird-Dominated Movement in Radar Images via ML — *Python, machine learning, image processing*
Working with the company Agrinerds to help manage the Highly Pathogenic Avian Influenza (HPAI) disease carried by waterfowl, a major cause of the depopulation of over 90 million poultry in North America across two outbreaks (2014–2015, 2022–present). Used NEXRAD radar images and image-processing-based machine learning to automate identifying which radar images showed bird-dominated movement.

Converter of Handwritten Document Images to LaTeX Files — *Python, Handwriting OCR*
Built a program that receives a picture of a handwritten math document and converts it into a LaTeX file of that document using Handwriting OCR and Python.

Virtual Settlers of Catan — *MATLAB App Designer, ThingSpeak*
Built an online version of Settlers of Catan using MATLAB App Designer in conjunction with ThingSpeak to make it online-compatible.

FPS-Style 1-on-1 Castle Attack and Defense Game — *Unreal Engine 5*
Developing a game similar to Clash Royale where the player is immersed in the fight with multiple shifting perspectives and detailed graphics: each player has a movable character fighting alongside their generated armies and defenses instead of a top-down view.

WORK EXPERIENCE

Sales Representative
Slimrate — POS Software, Hardware & Payments | August 2026 – Present (part-time, commission-based)

- B2B field sales of point-of-sale hardware, software, and payment processing to independent businesses in the Tracy, CA area: prospect local owners, qualify eligibility, and book discovery calls with the sales manager; building an automated weekly email-prospecting workflow to scale outreach.

High School Physics and Engineering Teacher, and Robotics Coach
Tracy Unified School District — Merrill F. West High School | August 2024 – June 2026

- Taught classes in the Space and Engineering Academy (SEA) at West High, overseeing students' additional academy responsibilities such as community service, senior projects, and other events.
- Classes taught: AP Physics C: Mechanics; AP Physics 1; Academy Physics (kinematics, forces, energy, optics, sound, electricity, nuclear physics, and more); Engineering Design (app, bridge, rocket, and robotics projects); Product Design (3D printing, laser cutting, and electronics for student-manufactured products).
- Advisor for the West High Robotics Club, capped at 48 students (8 teams of 6) competing in VEX V5 Robotics; in 2026, 4 teams reached the regional championships and the program raised over $25,000 through fundraisers and hosted tournaments.

Coding Reviewer for AI Training Data
Outlier | May 2024 – Present

- Working on projects focused on improving the coding abilities of AI LLMs.

Substitute Teacher
Tracy Unified School District | April 2024 – Present

Mathematics Instructor
Mathnasium — The Math Learning Center | January 2023 – Present

- Tutoring children and teenagers in various ranges of mathematics.

UC Davis Student Housing and Dining Services Caterer
Olive & Vine Catering | April 2022 – March 2023

- Planning and providing for all catering events for various UC Davis departments.

Instacart Grocery Deliverer
Instacart | April 2022 – March 2023

Warehouse Associate
Amazon | March 2021 – October 2021